

| 4(a)            | <b>A</b> neutrophil ; <b>A</b> polymorphonuclear, leucocyte / granulocyte<br><b>B</b> lymphocyte ;                                                                                                                                                                                                                                 |           | 2                                                             |        |       |                |        |                 |       |   |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------|--------|-------|----------------|--------|-----------------|-------|---|
| 4(b)(i)         | <table><tr><th>substance</th><th>concentration in lymph compared to the concentration in blood</th></tr><tr><td>oxygen</td><td>lower</td></tr><tr><td>carbon dioxide</td><td>higher</td></tr><tr><td>red blood cells</td><td>lower</td></tr></table> <p>three rows correct = 2 marks<br/>one row or two rows correct = 1 mark</p>  | substance | concentration in lymph compared to the concentration in blood | oxygen | lower | carbon dioxide | higher | red blood cells | lower | 2 |
| substance       | concentration in lymph compared to the concentration in blood                                                                                                                                                                                                                                                                      |           |                                                               |        |       |                |        |                 |       |   |
| oxygen          | lower                                                                                                                                                                                                                                                                                                                              |           |                                                               |        |       |                |        |                 |       |   |
| carbon dioxide  | higher                                                                                                                                                                                                                                                                                                                             |           |                                                               |        |       |                |        |                 |       |   |
| red blood cells | lower                                                                                                                                                                                                                                                                                                                              |           |                                                               |        |       |                |        |                 |       |   |
| 4(b)(ii)        | <i>two from</i><br>immune response occurs (linked to antibody / cytokine production) ;<br>increased concentration / production of / AW, of antibodies (which are protein) ;<br>AVP ; e.g. cytokine production                                                                                                                      | 2         |                                                               |        |       |                |        |                 |       |   |
| 4(c)(i)         | <i>two from</i><br>emits, impulses / waves of depolarisation / waves of excitation / AW ;<br>at regular intervals / AW ; e.g. acts as a pacemaker<br>spreads across the, atria / atrial muscles / AW<br><b>or</b><br>stimulates / AW, atrial, contraction / systole ;<br><b>A</b> initiates, heart beat / contraction of the heart | 2         |                                                               |        |       |                |        |                 |       |   |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 4(c)(ii) | <p><i>three from</i></p> <p>faster heart rate / increased rate of contraction (of ventricles) ;</p> <p>no, slowing / delay of, impulse / wave of excitation / AW, by AVN ;</p> <p>more impulses reaching the ventricle per unit time ;</p> <p>ventricles may contract before they are full ;</p> <p>irregular heart rate ;</p> <p>AVP ; e.g. may have no effect if only a small proportion of impulses pass through</p> <p>suggestion that ventricles may not contract from apex up</p> | 3     |